

Lab 3 — Written Architectural Justification

Assigned System: Community Solar Credit Allocation Manager (Problem Statement #63)

Architecture Selection: We chose **Layered Architecture** for the Community Solar Credit Allocation Manager.

Architectural Choice and Reasons

- **1. Clear separation of concerns:** The system can separate the Prosumer Portal/UI, business services (Order Manager and Payment Service), and data/integration services (Smart Meter Data Service and Solar Credit Database). This makes the credit-transfer and billing logic easier to maintain while keeping hardware/data concerns separate.
- **2. Suitable for the stated system scope:** The problem statement centers on ingesting solar-generation data, transferring credits between neighborhood accounts, and tracking monthly billing offsets. A layered structure provides a straightforward flow from user interaction to business processing and persistent data, matching these requirements without introducing the operational complexity of many independently deployed services.

Security Advantage

Layer boundaries allow access to sensitive credit, account, meter, and billing data to be controlled through defined service interfaces. Authentication/authorization can be enforced at the portal and business-service boundaries, while the database and meter-ingestion components remain behind those interfaces.

Performance Benefit

Separating presentation, business processing, and data access allows frequently requested credit/account information to be cached and database queries to be optimized without changing the user-interface layer. The meter-processing path can also validate and store hourly telemetry independently of the presentation logic.

Architecture Comparison

Style	Relevant characteristics for this scenario
Layered	Clear separation of concerns; straightforward data flow; suitable for traditional applications. Main trade-offs are layer overhead and
Microservices	Independent deployment/scaling and fault isolation. Trade-offs include operational complexity, network latency, and data-consisten
Client-Server	Centralized control and simpler consistency. Trade-offs include a possible single point of failure and scalability bottleneck.

Note: The Lab 3 handout requires the given Order Manager and Payment Service components. In this scenario, Order Manager represents credit-transfer orchestration, while Payment Service handles billing-offset/payment processing.